



MarketLink

Test Plan and Test Report

TechWiz 2026 · End-to-End Web Solutions

Team Omniverse · Aptech Learning Centre, F.B. Area, Karachi

Note for the team. This report was drafted by Claude Code (an AI coding assistant) from the real automated test runs listed in section 6. The SRS does not allow fully AI-made documentation, so please review it, repeat some of the manual checks yourselves, write the conclusions in your own words and keep the AI tool in your list of AI tools used.

1. Purpose and scope

This document describes how MarketLink was tested and what the results were. It covers every functional requirement of SRS section 1.6 (customer, farmer, admin and other features), the non-functional requirements of section 1.7 (responsive design, compatibility, security, usability) and the extra features added by the team (Urdu, the AI assistant in three languages, restock reminders, receipt confirmation, best sellers, the offer banner and others).

Out of scope, as in the SRS: online payment, delivery and verification of farmer licences.

2. Test approach

Level	What is tested	How
API tests	Every REST endpoint: correct answers, input checks, error messages, role checks (guest / customer / farmer / admin), rate limits	Node.js scripts that call the running server and compare the answers (PASS / FAIL per check)
Browser tests (end to end)	Complete user journeys in a real browser: register, log in, browse, filter, map, basket, checkout, orders, reviews, farmer and admin work	Playwright driving Chromium against the built app
Responsive tests	Every page at many screen widths in English and Urdu: nothing sticks out, no sideways scrolling	Playwright at widths from 320 to 1920 px
Regression tests	English text unchanged by the Urdu work; old features still working after each round of changes	The full set of suites run again after every round
Manual checks	Look and feel, screenshots of every page, e-mails in a local SMTP inbox	Screenshots reviewed page by page
Static checks	Code style and common mistakes	ESLint on the server and the client

Test data: the demo data made by `npm run seed` (project report, section 5). The database is seeded again before every suite, so each suite starts from the same data.

3. Test environment

Item	Value
Operating system	Linux (container), 4 CPU cores
Node.js	22

Item	Value
Database	MongoDB 8 (also run earlier against FerretDB, a MongoDB-compatible database)
Browser	Chromium (Playwright) at desktop, tablet and phone sizes, with touch for phones
Server	The production build (<code>npm run build</code> , <code>npm start</code>) on port 5000, time zone Asia/Karachi
E-mail	A local SMTP server that stores every e-mail, so e-mails can be checked without sending them
Maps	Map tiles replaced by a local image during tests (the tests do not depend on the internet)

4. Entry and exit criteria

- **Entry:** the build succeeds, ESLint reports no errors, the demo data loads.
- **Exit:** every automated check passes; any failure is either fixed in the code or, when the test itself expected old behaviour that was changed on purpose, the test is updated and the reason written down (section 7).

5. Test cases

Status: **Pass** means the case passed in the final run (section 6). The "Suite" column names the automated test that covers the case.

5.1 Customer features

ID	Requirement (SRS 1.6)	Steps	Test data	Expected result	Status	Suite
C-01	Customer registration	Open <code>/register</code> , fill every field, accept the terms, submit	New User, new@x.com, +92 300 1234567, Passw0rd	Account made, customer logged in and taken to the dashboard	Pass	smoke, e2e2
C-02	Registration checks	Submit with an empty name, a wrong e-mail, a short password, no terms	noterms@x.com and similar	Each is refused with a clear message; no account is made	Pass	smoke, e2e2
C-03	Phone number with country	Type a Pakistani number in the phone box	+92 321 5551234	Flag and dial code shown; saved as +923215551234	Pass	r10test, phone test
C-04	Login and logout	Log in with the demo customer, log out	customer@marketlink.com / Customer@123	Dashboard opens; after logout protected pages ask to log in	Pass	smoke, e2e
C-05	Wrong password / blocked account	Log in with a wrong password; with the deactivated customer	inactive.customer@marketlink.com	Refused with a message; blocked account cannot log in	Pass	smoke

ID	Requirement (SRS 1.6)	Steps	Test data	Expected result	Status	Suite
C-06	Forgot password	Ask for a reset link, open it, set a new password	customer@marketlink.com	Link e-mailed, valid 30 minutes, new password works, old one does not	Pass	fp
C-07	Favourite farmers and products	Press the heart on a farmer and a product, open Favourites	Malir Green Fields, Sindhri Mangoes	Both listed in Favourites; heart can be pressed again to remove	Pass	smoke, r4test
C-08	Family sharing (optional)	Add a family member, log in as the member	omar@marketlink.com	Member sees the shared household orders	Pass	smoke, api4
C-09	Browse markets by location and day	Open /markets , choose a day, a city, "near me"	Karachi, Saturday	Only matching markets; the page opens on the markets open today	Pass	audit3, r10test
C-10	Farmers at a market	Open a market page	Clifton Seaview Farmers Market	Farmers listed in a scroll box with "at the market today" status	Pass	r10test
C-11	Farmer profile	Open a farmer page	Malir Green Fields	Stall name, location, days, pickup windows, this week's stock, reviews	Pass	e2e, r4test
C-12	Map with markers and directions	Open /map , choose a market, ask for directions	Clifton market	Markers for markets and stalls; route drawn; Google Maps link	Pass	e2e2
C-13	Categories and filters	Filter products by category, price, market, day, rating and offers	Fruits, Rs 100 to 500	Only matching products; active filters shown as removable chips	Pass	smoke, r10test
C-14	Search while typing	Type in the shop search without pressing Enter; press X	"mango"	Results change after a short pause; X clears the box and the results	Pass	r10test
C-15	Navbar search suggestions	Type part of a word in the top search	"hon"	Suggestions (products, farmers, markets) appear after a short pause	Pass	r10test
C-16	Grid and list view	Switch the shop between grid and list		Products shown as cards or as rows	Pass	r10test
C-17	Product details	Open a product page	Sindhri Mangoes	Price, unit, quantity left, farmer, photos, reviews	Pass	smoke, r7test
C-18	Add to basket	Press Add, choose an amount, add	3 × a product	Basket count goes up; a toast offers "View basket"; the basket does not open by itself	Pass	r7test, r10test

ID	Requirement (SRS 1.6)	Steps	Test data	Expected result	Status	Suite
C-19	Pre-order with pickup slot	Check out, choose a pickup date and slot for each farmer	Next free slot	Order placed, stock reserved, confirmation page <code>/checkout/<order number></code> that can be bookmarked	Pass	e2e, r5test, r10test
C-20	Slot and stock checks	Order more than the stock, or after the cut-off	Quantity above stock	Refused with a message; nothing is reserved	Pass	smoke, audit3
C-21	Guest checkout	Check out without an account	A new e-mail	Account made, password e-mailed, toast "we sent your password to your email"	Pass	r5test
C-22	View, modify, cancel orders	Open an order, change the amount or slot, cancel	An order before its cut-off	Changes saved; cancelled order returns the stock; after the cut-off changes are refused	Pass	smoke, e2e
C-23	Order history and reorder	Open My orders, press "Order again"	A completed order	Past orders listed; items go back into the basket	Pass	smoke, r4test
C-24	Restock reminder	On a sold-out product press "Remind me" (as guest with an e-mail, and as customer)	A sold-out product	Reminder saved; refused for a product in stock; notification and e-mail when restocked	Pass	r10test
C-25	AI assistant	Ask about timings, farmers, products in English, Urdu and Roman Urdu	15 questions (section 5.4)	Correct answer from live data in the language asked	Pass	r10test, r8api
C-26	Receipt confirmation	After the farmer completes an order, answer "Did you receive your order?"	Demo customer's unconfirmed pickup	Yes → review form opens; No → farmer and admin told	Pass	r10test
C-27	Reviews and ratings	Rate a farmer and a product after pickup	4 stars and a comment	Review shown as a verified purchase; rating updated	Pass	e2e, r4test
C-28	See reviews before ordering	Open product and farmer pages		Reviews and average rating visible to everyone	Pass	r8test
C-29	Notifications	Place and update an order		In-app notification and e-mail for placed, accepted, ready, completed	Pass	smoke, e2e

5.2 Farmer features

ID	Requirement (SRS 1.6)	Steps	Test data	Expected result	Status	Suite
F-01	Farmer registration	Fill the sign-up wizard (account, stall, farm, markets, pickup days)	A new stall	Account made with status "pending"; selling features locked until approved	Pass	wizard, apitools, r4test
F-02	Profile with markets, days, windows and map pin	Edit profile, set a map pin, pickup windows and slot size	Latitude / longitude from the map	Saved and shown on the public profile and the map	Pass	e2e2
F-03	Add, edit, view, delete products	Manage products with photos	A new product with an image	Product listed in the shop; edits and deletes take effect	Pass	smoke, r4test
F-04	Weekly stock template	Set a recurring weekly amount	20 kg a week	Stock is refilled each week; history shows the template run	Pass	smoke
F-05	Sold out / unavailable	Mark a product sold out or hidden		Sold out shows "Remind me"; hidden products disappear from the shop	Pass	smoke, r4test, r10test
F-06	Low-stock alert	Let stock drop below the threshold	Threshold 5	Notification and e-mail to the farmer	Pass	api4
F-07	Handle pre-orders	Accept, decline (with a reason), mark ready, complete	Upcoming orders	Status changes; customer told each time; decline returns stock; buttons show a spinner while working	Pass	e2e, smoke
F-08	Cut-off time and slots	Change the cut-off hours and slot capacity	12 hours, 3 per slot	Checkout offers only allowed slots	Pass	audit3
F-09	Closed dates / not coming today	Add a closed date; press "I cannot come today"	Next Friday; today	Customers see it on the farmer and market pages; customers with a pickup are told	Pass	e2e2, r10test
F-10	Order history and insights	Open the dashboard and sales page		Total orders, pending orders, revenue, best-selling products, charts	Pass	api4, r4test
F-11	Reply to reviews	Write a reply to a review		Reply shown under the review	Pass	smoke, r8test

5.3 Admin and other features

ID	Requirement (SRS 1.6)	Steps	Test data	Expected result	Status	Suite
A-01	Separate admin dashboard	Log in as admin	admin@marketlink.com / Admin@123	Admin layout with totals of farmers, customers, markets, orders	Pass	adminui
A-02	Approve or suspend farmers	Approve the pending farmer; suspend one with a reason	Sunny Acres Poultry	Status changes; farmer e-mailed; customers of that market told about a new farmer	Pass	audit3, apitools

ID	Requirement (SRS 1.6)	Steps	Test data	Expected result	Status	Suite
A-03	Activate or deactivate customers	Deactivate and activate a customer		Deactivated customer cannot log in	Pass	smoke
A-04	Manage markets and cities	Add, edit, remove a market with map position	A new market	Market shown on the map and in lists	Pass	apitools
A-05	Content moderation	Remove a product listing and a review; handle a report	Reported review	Removed content disappears from the site	Pass	api4, r4test
A-06	Reports	Open the reports page, export CSV	Last 30 days	Total orders, revenue by market, most active farmers	Pass	api4, adminui
A-07	Farmer rankings	Rank farmers by revenue, rating, orders		Ranked table; admins only	Pass	r10test
A-08	Categories and announcements	Add a category; publish a seasonal announcement	Mango season, May to August	Category in filters; announcement shown only in its months	Pass	apitools, r6api
A-09	Offer banner	Change the percent, headline, photo and link; hide it	45%, "Up to {percent}% off summer mangoes"	Home page shows the new banner; outside links and percent over 90 refused; customers cannot change it	Pass	r10test
A-10	Contact messages by topic	Send a message with a topic; filter the admin inbox	Request a new market	Message saved with its topic; the filter shows only that topic	Pass	r10test
O-01	Role-based access	Call customer, farmer and admin endpoints with the wrong role or no login		401 without login, 403 with the wrong role	Pass	smoke, audit3
O-02	Search, sort, filter with map	Search markets, farmers, products; sort; use the map		Correct results and markers	Pass	smoke, e2e2
O-03	Responsive design	Open every page at many widths	320 to 1920 px	No sideways scrolling, nothing cut off; phone bottom bar on every page; dialogs as bottom sheets	Pass	resp8, r10test
O-04	Notifications on phones	Open the bell on a 360 px phone		Menu fits the screen	Pass	r10test
O-05	About us and contact us	Open /about and /contact		Team information; contact details with a map; topic chosen from the link	Pass	r3test, r10test
O-06	Urdu	Switch to Urdu on every public, customer and farmer page		All text in Urdu, layout mirrored right to left, no English left	Pass	r8test, resp8, r10test

ID	Requirement (SRS 1.6)	Steps	Test data	Expected result	Status	Suite
O-07	Photo credits	Check banners, product, farmer and market pages; open /credits		No credits on the photos; the credits page lists every photographer	Pass	r10test

5.4 AI assistant questions

ID	Question	Language	Expected answer	Status
AI-01	which farmer is highly rated	English	The highest-rated farmer (Malir Green Fields, 4.55 from 56 reviews) and a top-5 list	Pass
AI-02	sab se acha kisan kaunsa hai?	Roman Urdu	Same as AI-01	Pass
AI-03	سب سے اچھی ریٹنگ والا کسان کون ہے؟	Urdu	Same, written in Urdu	Pass
AI-04	tamatar kahan milega	Roman Urdu	Vine Tomatoes, price, farmer and pickup days	Pass
AI-05	sab se sasti sabzi	Roman Urdu	The five cheapest vegetables in stock	Pass
AI-06	zyada bikne wali cheezen	Roman Urdu	Top 5 best sellers with a link to the best sellers page	Pass
AI-07	kitne kisan hain	Roman Urdu	Number of approved farmers	Pass
AI-08	kaun se shehar mein mandi hai	Roman Urdu	Cities with their number of markets	Pass
AI-09	any offers this week?	English	Products on offer with the percent off	Pass
AI-10	markets open now	English	Markets open at this moment, or the ones opening later today	Pass
AI-11	what can I buy	English	Categories with the number of items in stock	Pass
AI-12	password bhoool gaya	Roman Urdu	How to reset the password, with a link	Pass
AI-13	I want to complain about a market	English	Contact page with the topics	Pass
AI-14	top rated fruits in karachi	English	Top-rated fruits sold at Karachi markets	Pass
AI-15	is Malir Green Fields at the market today?	English	Whether the farmer is there now, later today, has left, or is away	Pass
AI-16	how do I leave a review	English	The receipt question and the review form	Pass
AI-17	mera naam kya hai	Roman Urdu	Asks for the name instead of saving a wrong one	Pass

6. Test results

The whole set of suites was run on the final build (26 September 2026). The database was seeded again before every suite. After the fixes in section 7, every suite passes.

6.1 API tests

Suite	What it covers	Checks	Passed	Failed
smoke	Accounts, markets, farmers, products, basket rules, pre-orders, reviews, favourites, family, RBAC	69	69	0
smoke2	Filters, search, closed dates, reviews, maps data	23	23	0
audit3	SRS checklist: registration fields, near me, cut-off, slot capacity, suspend / approve	32	32	0
apitools	Admin tools: cities, markets, categories, accounts, place order, tables	33	33	0
api4	Inventory, low-stock alerts, sales insights, moderation, reports	31	31	0
r5api	Single login, guest checkout, verified reviews, SEO heads	25	25	0
r6api	Announcements by season, newsletter, FAQs, product SEO fields	44	44	0
r7api	FAQs admin, structured data, llms.txt, robots, sitemap	63	63	0
r8api	Urdu content in the API, server messages in Urdu, assistant memory	65	65	0
Total		385	385	0

6.2 Browser tests

Suite	What it covers	Checks	Result
e2e	Customer → farmer → admin journey (17 steps: order, accept, ready, complete, review, approval)	17 steps	Pass
e2e2	Map, route, city filter, closed dates, phone checks, redirects	6 steps	Pass
wizard	Farmer sign-up wizard, pending status	journey	Pass
fp	Forgot and reset password with the e-mail link	journey	Pass
r3test, avatartest, adminui	Footer, profile photo, admin shell, place order, CSV export	journeys	Pass
r4test	Password eye, shells, reviews, gallery, inventory, sales, moderation, phones	111	111 passed
r5test	Single login, dropdown search, quick view, basket sidebar, checkout without an account	45	45 passed
r6test	Banner, video, reviews, newsletter, seasonal announcements, DataTables	64	64 passed
r7test	Photos, 3 s banner, Add opens the amount dialog, empty basket, FAQs, SEO on the page	78	78 passed

Suite	What it covers	Checks	Result
r8test	Basket speed, reviews, Urdu on every page, a guest pre-order in Urdu	39	39 passed
r9test	Banner autoplay, toasts, category photos, offers, FAQ accordion, filter sidebar	63	63 passed
r10test	Round 10: assistant in 3 languages, offer banner, home layout, credits, reminders, receipt, away today, topics, rankings, live search, phones, Urdu	88	88 passed
dtsweep	Every DataTable at 1440 and 390 px	all tables	No problems
align, emoji, scan, sw	Alignment of every page, no emoji, no sideways scrolling at 360 / 390 / 768 / 1024 px when logged in	all pages	No problems (after fix 10)
resp8 (English and Urdu)	Every page at 18 widths, 320 to 1920 px	567 page views each	The only problems found were fixes 10 to 12; the affected pages were checked again after the fix

Total browser checks counted: 488 checks plus the step-by-step journeys and the sweeps.

6.3 Static checks

Check	Result
ESLint (server and client)	0 errors
Production build (<code>vite build</code>)	Succeeds
Missing Urdu translations (scan of every text in the client)	0 missing
Database script with validators + demo data	All 19 collections created; the full demo data loads without a validation error

7. Defects found and fixed

Serious or visible problems found by the tests during development, and how they were fixed:

#	Defect	Found by	Fix
1	Pages with a saved phone number (the customer profile) showed a blank page: the phone box asked for the full number before its number rules had loaded	r10test (bottom bar on the profile page)	The phone box uses the typed number until the rules are loaded
2	"Did you receive your order?" pop-up could cover the dashboard for every test user	Regression (older tests)	Expected behaviour for the demo customer; older tests skip that question, r10test checks it
3	The basket side panel shook when it opened	Manual check	The page keeps space for the scroll bar and the panel is focused without scrolling
4	The notification menu went off the left edge on phones	Responsive test	Full-width menu under the bell on phones

#	Defect	Found by	Fix
5	The assistant read "mera naam kya hai" as a new name ("What")	Assistant test	Roman Urdu phrase for "what's my name" added
6	"Top rated fruits in Karachi" was read as the farmer "Karachi Artisan Bakehouse"	Assistant test	A city name alone no longer picks a farmer
7	Map tiles were blocked (HTTP 403)	Manual check	Correct referrer policy and a second tile server as fallback
8	The home banner did not change by itself	Manual check (round 9)	Autoplay timer fixed; it pauses only with the pause button or keyboard focus
9	Farmer inventory page failed with a database error on an older database	API test against MongoDB	Query rewritten to work with the stored data
10	Customer area at 992 to 1199 px: the website links added to the top bar pushed the account menu 69 px off the screen	Responsive sweep (1024 px)	The links show from 1200 px; the sidebar has the same links below that
11	Markets page on phones (320 to 412 px): the Filters, Near me and view buttons ran off the left edge when the day filter was on	Responsive sweep	The buttons wrap onto a second line
12	Urdu top bar at 768 px was 6 px too wide because of the long Urdu "Sell with us" label	Responsive sweep (Urdu)	Slightly smaller padding for that button in Urdu at tablet widths
13	The new offer banner's label style also changed the "% off" tag on product pages (same class name)	r9test (product page offer)	The banner label has its own class
14	The toast title "Account created" after a guest checkout stayed in English on the Urdu site	r8test (guest pre-order in Urdu)	Urdu text added

Tests changed on purpose (the behaviour was changed at the customer's request, not a defect):

- Adding to the basket no longer opens the basket; the old tests now open it from the toast's "View basket" button.
- After checkout the address is `/checkout/<order number>` instead of `/checkout/success`.
- The markets page opens on the markets open today.
- The home page trust row, category heading and banner classes changed with the redesign.

8. Known limitations

- Tests ran in Chromium only; Firefox and Safari were not tested automatically. The site uses standard features, but a short manual check in those browsers is recommended before the final demo.
- Real e-mail delivery depends on the SMTP details in `server/.env`; the tests use a local SMTP server.
- Maps need an internet connection for the map tiles; directions use the free OSRM service.
- The AI assistant is rule-based; very unusual questions get a polite "I couldn't find an answer" with suggestions.

9. Conclusion

All 385 API checks and all browser suites pass on the final build, in English and in Urdu, from 320 to 1920 px wide. Every functional requirement of SRS section 1.6 has at least one passing test case (section 5), and the non-functional requirements (responsive design, security through role checks, usability) are covered by the sweeps. The testing found and fixed fourteen real problems during the project (section 7). The application is ready for the demonstration; the team should still do a short manual check in Firefox and Safari and send a real e-mail with their own SMTP details before submission.